

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		Indicative content $\text{RX} + \text{OH}^- \rightarrow \text{ROH} + \text{X}^-$ (any halogenopropane / NaOH) mechanism is nucleophilic substitution C—halogen bond must be broken for the reaction to take place based on electronegativity, carbon atom in the chlorocompound is the most δ^+ suggests that chlorocompound would react fastest based on bond enthalpies, the C—I bond is the weakest suggests that iodo compound would react fastest iodo compound does in fact react fastest so effect of bond strength effect is more significant than effect of polarity	1	1				
							4	6		6

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				5-6 marks Equation and nucleophilic substitution included; both bond strength and polarity considered and effects compared <i>The candidate constructs a relevant, coherent and logically structured account including key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary is used accurately throughout.</i> 3-4 marks Equation or nucleophilic substitution included; some reference to both bond strength and polarity <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Reference to the reaction type or to the data given <i>The candidate attempts to link relevant points from the indicative content. Coherence is limited by omission and/or inclusion of irrelevant material. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						
		(ii)		$\text{Ag}^+(\text{aq}) + \text{X}^-(\text{aq}) \rightarrow \text{AgX}(\text{s})$ accept any halide		1		1		
		(iii)		look down through solution / put a cross on paper underneath flask (1) record time taken for cross to be obscured (1) OR colorimeter (1) record time taken to reach set absorbance (1)		2		2		2
		(iv)		two layers formed / they would not have mixed / precipitate only formed at interface			1	1		1

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	(b)	(i)		CFCs produce chlorine radicals / $\text{Cl}\cdot$ (1) these destroy the ozone layer (1) the ozone layer protects us from UV radiation (1) UV radiation causes (skin) cancer / mutations (1) ignore all references to global warming	4			4		
		(ii)		C—F / C—H / C—C bonds would need to be broken to form radicals (1) these are too strong to be broken by UV radiation (1)		2		2		
				Question 10 total	5	6	5	16	0	9